11c. Task-Based Parallelism: @spawn

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INTRODUCTION

The previous section introduced the basics of multithreading. We highlighted that operations can be computed either sequentially (Julia's default) or concurrently. The latter enables the processing of multiple operations simultaneously, with each of them running as soon as a thread becomes available. When Julia's session is initialized with more than one thread, this determines that computations can be executed on different CPU cores in parallel.

The current section will focus on Julia's native multithreading mechanisms. The topic will span several sections. Our primary goal here is to demonstrate how to write multithreaded code, rather than exploring how and when to apply the technique. We've deliberately structured our explanation in this way to smooth subsequent discussions.

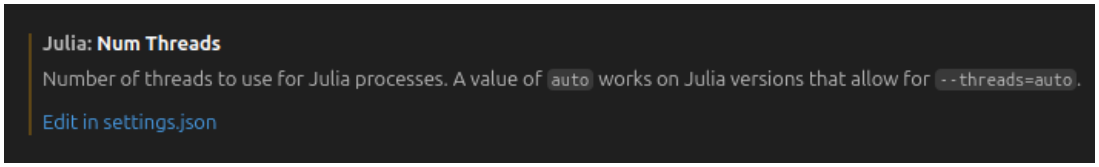
> ⚠️ **Warning!**
> While multithreading can offer significant performance advantages, it's not applicable in all scenarios. In particular, **multithreading demands extreme caution in handling dependencies between operations**, as mismanagement can lead to silent catastrophic bugs.
>
> This is why the section only aims at grasping a basic understanding of parallelism techniques. We'll devote an entire section on thread-unsafe operations.

ENABLING MULTITHREADING

Each Julia session is initialized with a given pool of threads available. Since each thread can run a set of instructions independently, the total number of threads determines how many instructions can be processed simultaneously.

By default, Julia runs in single-threaded mode. Consequently, you must configure it to enable multithreading. Importantly, **once this configuration is set, the change is permanent**: every new Julia session will start with the number of threads you specified.

In VSCode or VSCodium, multithreading is enabled by navigating to *File* > *Preferences* > *Settings*. Then, you must search for the keyword *threads*, which will display the following option:



After selecting *Edit in settings.json*, add the line `"julia.NumThreads": "auto"` in the file that opens. The specification `auto` automatically detects the number of threads supported by your system, typically matching the logical or physical cores of your CPU.

Notice the effects will only take place after a new session is started. To check whether the effects have taken place, run the command `Threads.nthreads()`. This displays the number of threads available in the running session. Any number greater than one will indicate multithreading is active.

Once a multithreaded session is started, there are several packages available for performing parallel computations. The current section will focus on the capabilities built directly into Julia, which are provided by the `Threads` package. This package is automatically "imported" in every Julia session, so you can call its functions and macros by prefixing them with `Threads`. Alternatively, you can load the package with `using`, which lets you use its macros and functions without qualification.

PACKAGE THREADS AUTOMATICALLY IMPORTED

```julia
# package `Threads` is automatically imported when you start a Julia session

Threads.nthreads()
```

```
24
```

USING THREADS

```julia
using Base.Threads       # or `using .Threads`

nthreads()
```

```
24
```

> ⚠️ **Warning!** - Loaded Package
>
> All the scripts below assume that you've executed `using Base.Threads`. This allows direct access to macros like `@spawn`.
>
> Furthermore, every example is based on a session running with **two worker threads**.

TASK-BASED PARALLELISM: @SPAWN

The first approach to parallel execution we introduce is the macro `@spawn`. By prefixing an expression with `@spawn`, Julia creates a (non-sticky) task that's immediately scheduled for execution. If a thread is available, the task begins running right away.

Unlike other parallel programming approaches we'll examine later, `@spawn` requires explicit instructions to wait for task completion. The method for doing so depends on the nature of the task output.

The `fetch` function is required when tasks return a result. It serves a dual purpose: it waits for a task to complete and also returns the task's output. Since parallel programs involve multiple spawned tasks, `fetch` should be broadcast over a collection containing all tasks.

The following example demonstrates `fetch` with two spawned tasks, each returning a vector.

BASELINE (NON-THREADED)

```
x = rand(10); y = rand(10)

function foo(x)
    a      = x .* -2
    b      = x .*  2

    a,b
end
```

FETCH

```
x = rand(10); y = rand(10)

function foo(x)
    task_a = @spawn x .* -2
    task_b = @spawn x .*  2

    a,b    = fetch.((task_a, task_b))
end
```

Note that `fetch` takes tasks as input. This is why it's important to distinguish between `task_a` and `a`:

- `task_a` denotes the task creating the vector `a` (i.e., the task itself),
- `a` refers to the vector created (i.e., the task's output).

Alternatively, for tasks that perform actions without returning a value, we can use either the function `wait` or the macro `@sync`. The function `wait` works analogously to `fetch`, except that it doesn't return any value. For its part, the macro `@sync` requires enclosing all the operations to be synchronized between the keywords `begin` and `end`.

To illustrate, consider a mutating function. Such functions are suitable examples for both cases, since they don't return a value.

BASELINE (NON-THREADED)

```
x = rand(10); y = rand(10)

function foo!(x,y)
    @. x = -x
    @. y = -y
end
```

WAIT

```
x = rand(10); y = rand(10)

function foo!(x,y)
    task_a = @spawn (@. x = -x)
    task_b = @spawn (@. y = -y)

    wait.((task_a, task_b))
end
```

@SYNC

```
x = rand(10); y = rand(10)

function foo!(x,y)
    @sync begin
        @spawn (@. x = -x)
        @spawn (@. y = -y)
    end
end
```

MULTITHREADING OVERHEAD

To see `@spawn` in action, let's calculate both the sum and the maximum of a vector `x`. Their computation will be implemented following a sequential and a simultaneous approach. These implementations will reveal that the total runtime of the sequential procedure is essentially the sum of the individual runtimes. In contrast, thanks to parallelism, the runtime under multithreading is roughly equivalent to the longer of the two computations.

JULIA'S DEFAULT (SEQUENTIAL)

```julia
x = rand(10_000_000)

function non_threaded(x)
    a           = maximum(x)
    b           = sum(x)

    all_outputs = (a,b)
end
```

```
julia> @btime maximum($x)
  7.479 ms (0 allocations: 0 bytes)
julia> @btime sum($x)
  2.986 ms (0 allocations: 0 bytes)
julia> @btime non_threaded($x)
  10.586 ms (0 allocations: 0 bytes)
```

MULTITHREADING

```julia
x = rand(10_000_000)

function multithreaded(x)
    task_a      = @spawn maximum(x)
    task_b      = @spawn sum(x)

    all_tasks   = (task_a, task_b)
    all_outputs = fetch.(all_tasks)
end
```

```
julia> @btime maximum($x)
  7.479 ms (0 allocations: 0 bytes)
julia> @btime sum($x)
  2.986 ms (0 allocations: 0 bytes)
julia> @btime multithreaded($x)
  7.849 ms (13 allocations: 1.031 KiB)
```

Although the multithreaded runtime is close to the maximum of the individual runtimes, the equivalence isn't exact. The reason is that **multithreading introduces overhead** due to the creation, scheduling, and synchronization of tasks. A practical consequence of this behavior is that **multithreading can actually be counterproductive for small workloads**, where the overhead can outweigh any performance gains.

To make this trade-off concrete, let's compare the execution times of a sequential and multithreaded approach for different sizes of `x`. In the examples, multithreading is only advantageous for sizes greater than 100,000, with the single-threaded approach dominating otherwise. The high threshold illustrates just how significant the overhead can be relative to the actual work performed.

NON-THREADED

```
x_small  = rand(    1_000)
x_medium = rand(  100_000)
x_big    = rand(1_000_000)

function foo(x)
    a           = maximum(x)
    b           = sum(x)

    all_outputs = (a,b)
end
```

```
julia> @btime foo($x_small)
  878.406 ns (0 allocations: 0 bytes)
julia> @btime foo($x_medium)
  58.661 μs (0 allocations: 0 bytes)
julia> @btime foo($x_big)
  649.520 μs (0 allocations: 0 bytes)
```

MULTITHREADED

```
x_small  = rand(    1_000)
x_medium = rand(  100_000)
x_big    = rand(1_000_000)

function foo(x)
    task_a      = @spawn maximum(x)
    task_b      = @spawn sum(x)

    all_tasks   = (task_a, task_b)
    all_outputs = fetch.(all_tasks)
end
```

```
julia> @btime foo($x_small)
  2.245 μs (13 allocations: 1.031 KiB)
julia> @btime foo($x_medium)
  62.828 μs (13 allocations: 1.031 KiB)
julia> @btime foo($x_big)
  572.777 μs (13 allocations: 1.031 KiB)
```